



Research Configuration

Deterministic Seed Initialization
(PrngSeedGenerator)

ADBG Synthetic Academic Document Generation

Typst Vector Ground Truth PDF from JSON Assembly

Raster Image Conversion (300 DPI)

Controlled Degradation Matrix
(clean | scanner | camera | rotated)

AU_DIC_Benchmark_v1.0 Dataset Assembly
(N = 360 Specimen Suite)

Benchmark Runner Engine Initialization
(isReadOnly: true, allowMockFallback: false)

Live Vision-Language / OCR Model Inference
(Groq Cloud Llama 3.1 8B Instant)

Raw Prediction Extraction & JSON Parsing

Six-Stage Semantic Canonical Normalization
(CanonicalNormalizer)

Field-Level Candidate Comparison

Nine-Class Structured OCR Error Taxonomy
(ErrorTaxonomist)

Multi-Task Metric Computation

Statistical Significance & Uncertainty Analysis

Publication Artifact Generation